Introduction to Artificial Intelligence

Sample knowledge-base document for RAG ingestion and retrieval testing.

1. What Is Artificial Intelligence?

Artificial intelligence is the field of building computer systems that perform tasks associated with capabilities such as perception, reasoning, learning, language understanding, planning, and decision-making.

Key idea: What Is Artificial Intelligence? is an important foundational topic when learning and applying introduction to artificial intelligence. Understanding the concept, its purpose, and common examples helps when building practical software systems.

2. AI and Machine Learning

Machine learning is a major approach within AI. Traditional AI can also include rule-based systems, search algorithms, planning, optimization, and knowledge representation. Machine learning allows systems to learn patterns from examples.

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3. Deep Learning

Deep learning is a subset of machine learning based on neural networks with multiple computational layers. It has achieved strong results in computer vision, speech recognition, language processing, and many other areas.

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4. Natural Language Processing

Natural language processing focuses on enabling computers to work with human language. Tasks include classification, translation, information extraction, summarization, question answering, and text generation.

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5. Large Language Models

Large language models are neural network models trained on large amounts of text and related data. They can generate and transform language, answer questions, summarize information, write code, and support many application workflows.

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6. Generative AI

Generative AI systems create new content such as text, images, audio, video, or code. A generative model learns patterns in training data and produces outputs based on an input prompt or other conditioning information.

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7. Embeddings

An embedding represents an item such as text as a numerical vector. Similar meanings can be represented by vectors that are close under an appropriate similarity measure. Embeddings are commonly used for semantic search and retrieval.

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8. Retrieval-Augmented Generation

RAG combines information retrieval with generation. Documents are processed into chunks, converted into embeddings, and stored in a searchable system. A user question is embedded and relevant chunks are retrieved before the language model generates an answer.

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9. AI Agents

An AI agent is a system that can use a model together with tools, state, and workflows to accomplish a goal. An agent may retrieve information, call APIs, inspect files, make decisions, and perform multiple steps.

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10. Responsible AI

AI systems should be designed with attention to accuracy, privacy, security, fairness, transparency, robustness, and human oversight. Evaluation should measure whether a system performs reliably for the intended users and tasks.

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